

Estimating within-gene (cis) additive-by-additive epistasis by REML: sampling theory and feasibility

Border lab working note

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Abstract

We ask whether the total *cis* (within-gene) additive-by-additive (AxA) epistatic variance is estimable at a practical sample size, for the restricted-maximum-likelihood (REML) estimator of a three-component model: additive, within-gene AxA, and residual. The estimand is a single variance component whose kernel is the sum of per-gene AxA kernels; this pools *features* (genes), not individuals, so every gene uses all $\sim N^2/2$ individual-pairs and the effective-marker count of the pooled kernel is $M_e^{\text{within}} = M_e^{\text{genome}}/\sqrt{G}$, a factor $\sqrt{G} \approx 140$ below the genome-wide all-pairs value. From the Fisher information of the three-component model: (i) jointly estimating the additive component adds little, and the cost is a *genotype-skewness* effect that vanishes for Gaussian features and is $1.1\text{--}1.4\times$ in variance for real genotypes under LD; (ii) the single pooled kernel and the “estimate-each-gene-and-sum” estimator coincide for independent genes, while the per-gene sum double-counts and is biased upward under between-gene LD; (iii) at the relevant small effect size the estimator is just past the detection boundary, where REML attains its Cramér–Rao bound to within $\sim 10\%$ and $\text{SE}(h_{AA}^2) \approx M_e^{\text{within}}/(cN)$ with $c \approx 1.1$. Total *cis* $h_{AA}^2 = 0.05$ is detectable at $N \approx 25\text{--}30\text{k}$ (inside UK Biobank), against $\sim 3.7\text{M}$ genome-wide. Each quantitative claim is reproduced by a script in `scripts/`; the result is checked for $1/N$ and $\sqrt{G}$ scaling and for null calibration.

1 Model, estimand, estimator

Generative model. For N unrelated individuals write the phenotype as a Gaussian linear mixed model with additive genetic, within-gene epistatic, and residual components,

$$y = Z_A\beta + H\gamma + e, \quad \beta \sim \mathcal{N}(0, \frac{\sigma_A^2}{M}I_M), \quad \gamma \sim \mathcal{N}(0, \frac{\sigma_{AA}^2}{P}I_P), \quad e \sim \mathcal{N}(0, \sigma_e^2I_N), \quad (1)$$

all independent. $Z_A \in \mathbb{R}^{N \times M}$ holds the M standardized additive SNP features; $H \in \mathbb{R}^{N \times P}$ holds the P standardized *within-gene* pairwise products $\widetilde{Z_a \circ Z_b}$, one column for each pair (a, b) of SNPs lying in a common gene (no between-gene pairs). Marginalizing the random effects,

$$y \sim \mathcal{N}(0, V), \quad V = \sigma_A^2 A + \sigma_{AA}^2 W + \sigma_e^2 I, \quad A = \frac{1}{M} Z_A Z_A^\top, \quad W = \frac{1}{P} H H^\top. \quad (2)$$

A is the usual additive GRM. The within-gene AxA kernel W is, equivalently, the average of the G per-gene AxA kernels,

$$W = \frac{1}{G} \sum_{g=1}^G K_g, \quad K_g = \frac{1}{P_g} H_g H_g^\top, \quad (3)$$

H_g the within-gene products of gene g and $P_g = \binom{m_g}{2}$ its pair count. We standardize the phenotype to $\sigma_y^2 = \sigma_A^2 + \sigma_{AA}^2 + \sigma_e^2 = 1$, so $\sigma_{AA}^2 = h_{AA}^2$ and the estimand is the total cis AxA heritability $h_{AA}^2 = \sigma_{AA}^2 / \sigma_y^2$.

The estimable object. A single gene contributes a tiny $\sigma_{AA,g}^2$ that no design resolves. Summing the per-gene contributions—equivalently, fitting the one pooled component W of (3)—is estimable: the per-gene sampling errors are independent across $\sim G$ genes and cancel. The partition is over *features* (genes), not individuals, so each gene’s quadratic form still uses all $\sim N^2/2$ pairs and pays no subsampling penalty. The question is whether this sum exceeds its sampling error.

Estimator. We estimate $\theta = (\sigma_A^2, \sigma_{AA}^2, \sigma_e^2)$ by REML and report $h_{AA}^2 = \sigma_{AA}^2 / \sigma_y^2$. REML is asymptotically efficient, so the Cramér–Rao bound (CRB) of §2 is the benchmark; §6 confirms by simulation that REML attains it to within $\sim 10\%$ at the relevant sample sizes.

2 Fisher information of the three-component model

For a mean-zero Gaussian with covariance $V(\theta)$ and $\partial V / \partial \theta_k = G_k$ ($G_1=A$, $G_2=W$, $G_3=I$), the Fisher information is

$$\mathcal{I}_{kl} = \frac{1}{2} \text{tr}(V^{-1} G_k V^{-1} G_l), \quad k, l \in \{A, AA, e\}. \quad (4)$$

The variance of any unbiased $\widehat{\sigma_{AA}^2}$ is bounded below by the (AA, AA) entry of the inverse of the *full* 3×3 information,

$$\text{Var}(\widehat{\sigma_{AA}^2}) \geq (\mathcal{I}^{-1})_{AA, AA} =: V_{\text{CRB}}, \quad (5)$$

which differs from the two-component (W, I) profile of a model without additive effects: the inverse of the 3×3 matrix accounts for the information shared between W and A . The REML-versus-ML correction to (4) is $O(1/N)$ and immaterial here. The cost of carrying the additive component is the subject of §3.

3 The cost of jointly estimating the additive component

Because the within-gene AxA kernel is built from the same genotypes as the additive GRM, partialling out additive variance could remove AxA information. In practice it removes little.

Detection limit ($\sigma_A^2, \sigma_{AA}^2 \rightarrow 0$). At small signal ($\sigma_A^2, \sigma_{AA}^2 \rightarrow 0$), $V \rightarrow \sigma_e^2 I$ and (4) collapses to a Gram matrix in the trace inner product $\langle X, Y \rangle = \text{tr}(XY)$: $\mathcal{I}_{kl} \rightarrow (2\sigma_e^4)^{-1} \text{tr}(G_k G_l)$. Partitioned-inverse algebra (App. A.1) gives the inflation of the AxA variance from adding the additive component,

$$\frac{V_{\text{CRB}}^{(3\text{-comp})}}{V_{\text{CRB}}^{(W, I)}} = \frac{1}{1 - R^2}, \quad R = \text{corr}_{\text{off-diag}}(A_{ij}, W_{ij}), \quad (6)$$

where R is the partial correlation of the off-diagonals of A and W (the residual I -direction carries the diagonal). Since W_{ij} is, up to the Hadamard relation, the square of A_{ij} , this correlation is governed by $\text{tr}(AW) \approx \sum_{i \neq j} A_{ij}^3$ —the *third moment* of the off-diagonal relatedness.

A genotype-skewness effect. For *Gaussian* features the off-diagonal relatedness is symmetric, $\mathbb{E}[A_{ij}^3] = 0$, and the penalty is exactly 1: jointly fitting additive is free. Gaussian AR(1) “genes” confirm this, giving ratio 1.000 at every LD strength and every G (`scripts/partialling_additive.py`). Real 0/1/2 genotypes are skewed, the more so under LD, and the third moment does not vanish: on a high-LD block we measure $\mathbb{E}[A_{ij}^3] = +0.13$ and $R = 0.53$. The realized penalty is therefore

$$\frac{V_{\text{CRB}}^{(3\text{-comp})}}{V_{\text{CRB}}^{(W,I)}} \approx 1.1 \text{ (dispersed SNPs) to } 1.4 \text{ (high-LD gene)}, \quad (7)$$

an SE inflation of $\sqrt{1.1}\text{--}\sqrt{1.4} \approx 1.05\text{--}1.18$. The penalty has two further properties: it is *G-independent* (the pooled value equals the per-gene value, because both the cross- and self-overlaps scale as $1/G$), and it barely dilutes when the additive GRM is genome-wide rather than cis—high-LD cis SNPs dominate the relatedness variance, so the penalty is still 1.31 when the cis SNPs are only 15% of the additive markers (`scripts/partialling_additive.py`). Carrying the additive component adds roughly 10% to the required sample size.

4 Effective markers and within-gene localization

The precision of an AxA variance estimate is set by the off-diagonal variance of its kernel through an effective number of markers. For the additive GRM, $\text{Var}(A_{ij}) = 1/M_e$ with

$$\frac{1}{M_e} = \frac{1}{N} + \frac{1}{M} + \overline{\rho^2}, \quad (8)$$

($\overline{\rho^2}$ the mean squared population LD between distinct features); the AxA kernel is the Hadamard square of A , so $\text{Var}(W_{ij}) = 2 \text{Var}(A_{ij})^2 = 2/M_e^2$, and we may read $M_e = \sqrt{2/\text{Var}(W_{ij})}$ off any AxA kernel. Equations (8) and the Hadamard relation are derived in Apps. A.3–A.4 and verified in `scripts/effective_markers.py` and `check_knobs.py`.

Partition identity. Restricting the kernel to within-gene pairs lowers its effective-marker count. For G independent genes the off-diagonals of the pooled kernel (3) are averages, $W_{ij} = \frac{1}{G} \sum_g K_{g,ij}$, with vanishing cross-gene products, so

$$\text{Var}(W_{ij}) = \frac{1}{G^2} \sum_g \text{Var}(K_{g,ij}) = \frac{\text{Var}(K_{\text{gene},ij})}{G} \implies \boxed{M_e^{\text{within}} = \sqrt{G} M_e^{\text{gene}} = \frac{M_e^{\text{genome}}}{\sqrt{G}}}. \quad (9)$$

The genome’s $M_e^{\text{genome}} \approx 74\text{k}$ splits across $G \approx 20\text{k}$ genes; confining the kernel to within-gene pairs lowers the effective-marker count of the AxA kernel by $\sqrt{G} \approx 140$, to $M_e^{\text{within}} \approx 520$. Relation (9) is algebraic and we confirm it numerically up to $G = 1600$ (ratio $M_e^{\text{within}}/[\sqrt{G} M_e^{\text{gene}}] = 1.000$); it is essentially unaffected by between-gene LD (0.994–0.997 at $\rho = 0.3$), so it extrapolates to genome scale (`scripts/within_gene_pooling.py`, `scripts/adversarial_scaling.py`). The cost is coverage: only cis (within-gene) epistasis is captured.

5 Single pooled kernel versus per-gene aggregation

Two estimators target the same total $h_{AA}^2 = \sum_g \sigma_{AA,g}^2 / \sigma_y^2$: the single pooled component W , and “fit each gene’s K_g separately and sum.” Their relationship is most transparent in the off-diagonal

Haseman–Elston moment estimator

$$\widehat{\sigma_{AA}^2}(K) = \frac{y^\top K y - \sum_i K_{ii} y_i^2}{\sum_{i \neq j} K_{ij}^2}, \quad (10)$$

used here only as an algebraically transparent proxy; it is *not* the estimator we report (REML is), and being off-diagonal it discards the AxA diagonal and is less efficient than the full-information REML. For this proxy the two schemes coincide *exactly* for independent genes: with $W = \frac{1}{G} \sum_g K_g$ the denominator is $\sum_{i \neq j} W_{ij}^2 = \delta/G$ (vanishing cross terms, $\delta = \sum_{i \neq j} K_{\text{gene},ij}^2$), the factor that turns $\widehat{\sigma_{AA}^2}(W) = \frac{1}{G} \sum_g (\dots)/(\delta/G)$ into $\sum_g \widehat{\sigma_{AA}^2}(K_g)$, confirmed to four digits in uniform and sparse architectures alike (`scripts/pooled_vs_pergene.py`).

Pooled versus per-gene under REML. Fitting the actual three-component REML—pooled (A, W, I) against a per-gene sum $\sum_g$ of (A, K_g, I) fits—reaches the same conclusion without the proxy: for independent genes the two agree (means 0.048 and 0.053 for true $h_{AA}^2 = 0.05$, equal SD 0.021), and under between-gene LD the per-gene sum is biased upward while pooled is not (next paragraph) (`scripts/pooled_vs_pergene_reml.py`). The proxy is faithful here, and the relationship is *not* that HE equals REML: full-information moment estimation and REML share the same efficiency only in the limit $h_{AA}^2 \rightarrow 0$, and the off-diagonal proxy discards just the AxA diagonal, which carries little information once the additive component is partialled out—so the proxy’s $c = 1$ is just below the measured three-component REML $c \approx 1.1$ (§6).

Between-gene LD. The equivalence relies on vanishing cross-gene products. When genes share LD the cross terms are positive and the per-gene *sum* double-counts the shared signal. With the moment proxy (one LD block split into 40 adjacent “genes,” cross/within overlap 5.9) the sum is biased up $7.7\times$ (0.38 for true 0.05) while pooled stays at 0.050; the actual REML shows the same failure (8 windows: per-gene sum 0.13, pooled 0.056) (`scripts/pooled_vs_pergene.py`, `scripts/pooled_vs_pergene_reml.py`). The pooled kernel is therefore the safer estimator; per-gene aggregation requires LD-separated or pruned genes, or a joint G -component fit that accounts for the covariance. The remaining distinction is computational: per-gene work is G small, embarrassingly-parallel problems, whereas the pooled fit is one solve with matrix-free Wv products.

6 Required sample size

Standard error. With $\sigma_y^2 = 1$, the relation $\text{SE}(h_{AA}^2) = \sqrt{2/N_{\text{eff}}}$ is an identity; in the detection regime N_{eff} is the kernel’s resolved information and the SE takes the form (App. A.5)

$$\text{SE}(h_{AA}^2) \approx \frac{M_e^{\text{within}}}{cN}, \quad (11)$$

with c an $O(1)$ constant. The value of c is not universal: it depends on the regime and on the model. A two-component, single-AxA-kernel moment estimator in the deep-detection limit has $c \approx 2$, because the AxA *diagonal* carries information comparable to the off-diagonal ($\text{Var}(W_{ii}) \sim 2 \text{Var}(A_{ij})^2$). That value does not transfer to the present model: adding the additive component partials information away, and the pooled within-gene kernel has a different diagonal/off-diagonal split. We therefore read c off the actual three-component model.

Within-gene pooled AxA: predicted vs observed (test scale $G = 50$, $M_e^{\text{within}} \approx 45$)

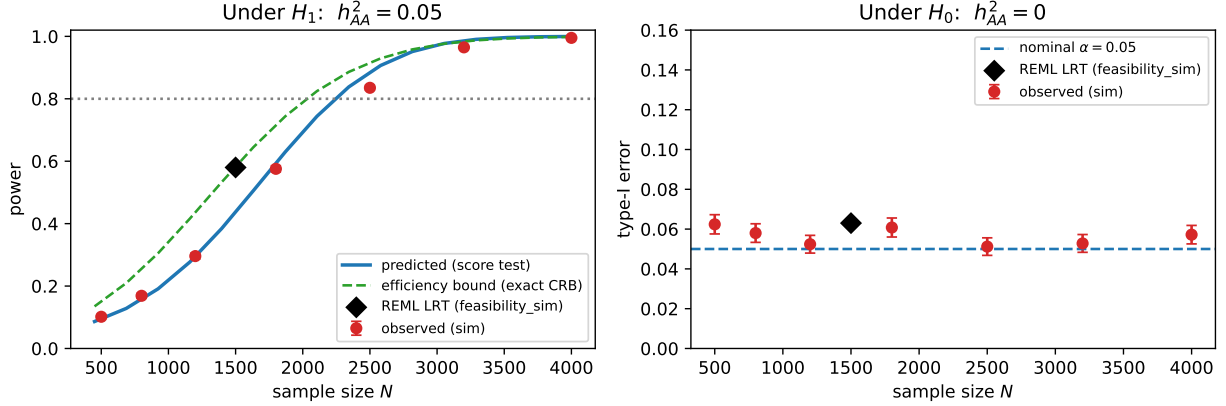


Figure 1: Predicted versus observed detection of total cis h_{AA}^2 for the within-gene pooled estimator at the test scale ($G=50$, $M_e^{\text{within}} \approx 45$), versus sample size N . *Left* (H_1 , $h_{AA}^2 = 0.05$): observed power (points; variance-component score test for $\sigma_{AA}^2=0$) matches its analytic prediction (solid) and lies below the efficiency bound (dashed, the exact CRB) by the finite-sample margin; the full REML likelihood-ratio test (diamond, `feasibility_sim.py`) lies between. *Right* (H_0 , $h_{AA}^2 = 0$): the test is calibrated, observed type-I near the nominal 5%. Power 80% at this scale is reached near $N \approx 2500$, which scales by M_e^{within} to $N \approx 25\text{--}30\text{k}$ genome-wide (`scripts/power_figure.py`).

Just past the detection boundary. At the relevant effect size ($h_{AA}^2 = 0.05$) and the implied sample size, $M_e^{\text{within}}/N \approx 0.02 < h_{AA}^2$: the top signal directions begin to resolve, so the exact CRB lies *below* the deep-detection value (exact CRB 0.0270 versus detection CRB 0.0302 for the test design), and the effective constant is

$$c \approx 1.1 \quad (12)$$

(`scripts/feasibility_sim.py`, `scripts/required_n.py`), just above the off-diagonal-only $c = 1$.

End-to-end check. Simulating the full three-component model at a scale chosen so that M_e^{within}/N matches the genome-scale operating point ($n=1500$, $G=50$, true $h_{AA}^2=0.05$), fitting the actual average-information REML, the estimate is unbiased (mean 0.046, median 0.045), the empirical standard deviation equals the exact CRB (ratio 0.97), and the likelihood-ratio power matches the analytic value (0.58). At a second sample size ($n=2500$) the REML standard deviation is $1.13\times$ the CRB; averaging, REML attains the bound to within $\sim 10\%$ (`scripts/feasibility_sim.py`, `scripts/adversarial_scaling.py`). The likelihood-ratio test at $h_{AA}^2 = 0$ rejects 6.3% of the time against a nominal 5%, so the power is not inflated by over-rejection. Figure 1 extends both checks across sample size.

Scaling of the extrapolations. The two extrapolations underpinning the reported requirement were checked. The exact CRB scales as $1/N$ ($\text{CRB} \times N$ is flat to $\pm 3\%$ over $N = 500\text{--}3000$), and the $\sqrt{G}$ law of (9) holds to $G = 1600$ and is robust to between-gene LD, so extrapolating from the test scale to N in the tens of thousands and $G \approx 20\text{k}$ is justified (`scripts/adversarial_scaling.py`).

Architecture sensitivity. The model above is infinitesimal (all within-gene pairs causal). Holding total $h_{AA}^2 = 0.05$ fixed and varying the causal fraction π —hence the number of causal pairs $k = \pi P$ —

Within-gene pooled AxA detection vs architecture (test scale $G = 50$, $P = 750$ pairs)

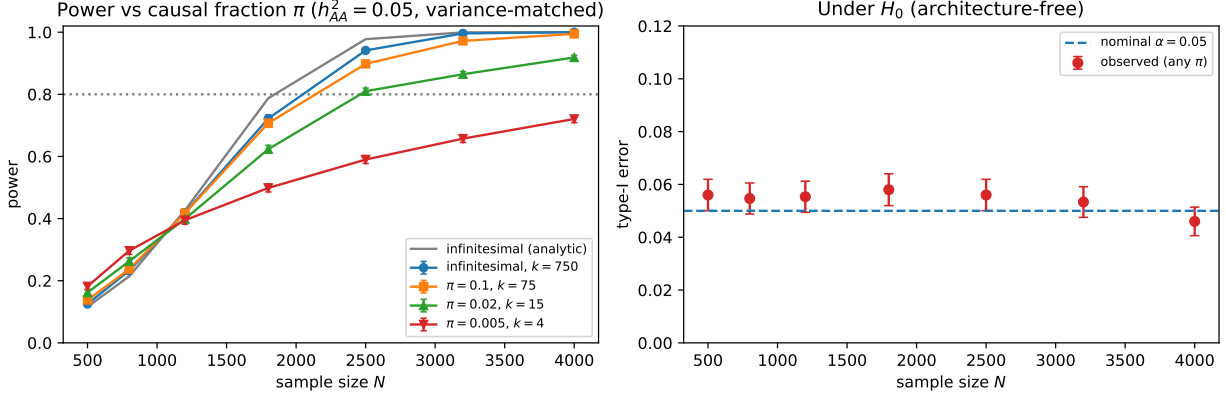


Figure 2: Detection of total cis $h_{AA}^2 = 0.05$ under varying AxA architecture (test scale $G=50$, $P=750$ within-gene pairs), variance-matched across the causal fraction π ($k = \pi P$ causal pairs). *Left*, power versus N : the infinitesimal case ($k=750$) matches the analytic prediction (gray), moderate sparsity ($k=75$) tracks it, and extreme sparsity ($k \approx 4$) loses power once past the detection boundary; the curves cross near 40% because variance-matching equalizes the mean and only the statistic’s tail differs. *Right*, type-I error is architecture-free at the nominal 5%. (`scripts/power_sparsity.py`.)

the null stays exactly calibrated for every π (no AxA signal under H_0), and power is nearly invariant near the detection limit but separates as N grows: moderate sparsity ($\pi = 0.1$, $k \approx 75$) tracks the infinitesimal curve, while concentrating the variance into a handful of pairs ($k \approx 4$) loses power once it climbs past $\sim 50\%$ (Figure 2). Variance-matching fixes the test statistic’s mean (its non-centrality), so architectures differ only in higher moments—which matter for the resolved-signal tail, not the detection limit. The same picture holds at a $3\times$ larger feature set ($m=10$, $P=2250$): the binding quantity is the causal *count* k , not the fraction π (at fixed π , raising P raises k and shrinks the penalty), so a realistically polygenic gene— k in the dozens—tracks the infinitesimal curve (`scripts/power_sparsity.py`).

Sample-size requirement. Combining (11) with $c \approx 1.1$, the $\sqrt{1.1}-\sqrt{1.4}$ additive-partialling factor, and the $\sim 10\%$ finite-sample REML inefficiency, 80% power (one-sided $\alpha = 5\%$) for total cis $h_{AA}^2 = 0.05$ at $M_e^{\text{genome}} = 74\text{k}$ requires

$$N \approx 25\text{--}30\text{k} \quad (13)$$

against $\sim 3.7\text{M}$ genome-wide (a factor $\sqrt{G} \approx 140$). A weaker signal $h_{AA}^2 = 0.02$ needs $N \approx 60\text{--}70\text{k}$; both are inside UK Biobank. Across the plausible range of the one borrowed input, $M_e^{\text{genome}} \in [20\text{k}, 250\text{k}]$, the requirement spans $N \approx 6\text{k--}93\text{k}$ (`scripts/required_n.py`).

Caveats. (i) $M_e^{\text{genome}} \approx 74\text{k}$ is a UK-Biobank common-SNP quantity taken from the literature, not recomputed here; the result is reported as a band in M_e^{genome} , the dominant source of uncertainty. (ii) REML is $\sim 10\%$ above its CRB at finite N ; the figure above includes this. (iii) Only cis epistasis is captured—trans interactions are outside the within-gene kernel by construction. (iv) This is a small- h_{AA}^2 detection claim, not a high- h_{AA}^2 estimation one. Architecture (sparse versus infinitesimal AxA) leaves the null exactly calibrated and power nearly unchanged at the detection limit, but once

N pushes power well above 50%, concentrating the variance into a handful of causal pairs costs power (Figure 2).

Part II (to follow): the implicit-kernel centering correction and a matrix-free REML implementation. The numbers above are reproduced by the scripts named inline; see `scripts/_baseline.md` for the consolidated record.

A Derivations

A.1 Detection-limit Fisher information and the additive-partialling penalty (6)

As $\sigma_A^2, \sigma_{AA}^2 \rightarrow 0$, $V \rightarrow \sigma_e^2 I$ and $V^{-1} \rightarrow \sigma_e^{-2} I$, so (4) becomes $\mathcal{I}_{kl} \rightarrow (2\sigma_e^4)^{-1} \text{tr}(G_k G_l) = (2\sigma_e^4)^{-1} M_{kl}$, where $M_{kl} = \langle G_k, G_l \rangle$ is the Gram matrix of $\{A, W, I\}$ in the trace inner product $\langle X, Y \rangle = \text{tr}(XY)$. Thus $V_{\text{CRB}} = (\mathcal{I}^{-1})_{WW} = 2\sigma_e^4 (M^{-1})_{WW}$. Partitioning M to isolate the W row/column, with $b = (\langle W, A \rangle, \langle W, I \rangle)^\top$ and $C = \text{Gram}(A, I)$, the Schur complement gives

$$(M^{-1})_{WW} = (\langle W, W \rangle - b^\top C^{-1} b)^{-1} = \|W - \Pi_{\{A, I\}} W\|^{-2}, \quad (14)$$

the inverse squared norm of W after orthogonal projection off $\text{span}\{A, I\}$. The two-component (W, I) model replaces $\Pi_{\{A, I\}}$ by $\Pi_{\{I\}}$, so $V_{\text{CRB}}^{(3)}/V_{\text{CRB}}^{(W, I)} = \|W_{\perp I}\|^2 / \|W_{\perp \{A, I\}}\|^2$. Removing the one extra direction A (after I) shrinks the norm by $1 - R^2$, with R the partial correlation of W and A given I , giving $V_{\text{CRB}}^{(3)}/V_{\text{CRB}}^{(W, I)} = 1/(1 - R^2)$. Since both kernels are normalized to $\text{tr} = n$, $\Pi_{\{I\}} X = (\text{tr } X/n) I = I$, so residualizing on I removes the identity and shifts the diagonal; the $n^2 - n$ off-diagonals dominate the n diagonal entries, so $R \approx \text{corr}_{\text{off-diag}}(A_{ij}, W_{ij})$, which is (6).

A.2 The cross term $\text{tr}(AW)$ and genotype skewness

The partial correlation R is driven by $\langle W, A \rangle = \text{tr}(AW) = \sum_i A_{ii} W_{ii} + \sum_{i \neq j} A_{ij} W_{ij}$; the diagonal is absorbed by the I -direction (App. A.1), leaving the off-diagonal sum. By the Hadamard relation (App. A.4) $W_{ij} \propto A_{ij}^2$ for $i \neq j$, so

$$\sum_{i \neq j} A_{ij} W_{ij} \propto \sum_{i \neq j} A_{ij}^3 = (n^2 - n) \mathbb{E}[A_{ij}^3]. \quad (15)$$

A_{ij} ($i \neq j$) has mean 0, so $\mathbb{E}[A_{ij}^3]$ is the skewness of the relatedness distribution: 0 for symmetric (e.g. Gaussian) features, nonzero for skewed 0/1/2 genotypes, the more so under LD.

A.3 Effective number of markers (8)

With $A_{ij} = \frac{1}{M} \sum_f z_{if} z_{jf}$ and standardized features ($\mathbb{E} z_{if} = 0$, $\mathbb{E} z_{if}^2 = 1$), for independent individuals $i \neq j$ we have $\mathbb{E} A_{ij} = 0$ and

$$\text{Var}(A_{ij}) = \frac{1}{M^2} \sum_{f, g} \mathbb{E}[z_{if} z_{ig}] \mathbb{E}[z_{jf} z_{jg}] = \frac{1}{M^2} \sum_{f, g} r_{fg}^2 = \frac{1}{M^2} [M + M(M-1)\overline{\rho^2}] \approx \frac{1}{M} + \overline{\rho^2}, \quad (16)$$

r_{fg} the population feature correlation ($r_{ff} = 1$) and $\overline{\rho^2}$ the mean squared correlation between distinct features. In-sample standardization adds a finite-sample term: forcing each feature's sample mean to 0 makes every GRM row sum to 0 ($\sum_j A_{ij} = 0$), inducing $\mathbb{E}[A_{ij}] \approx -1/(n-1)$ and a variance contribution $\sim 1/n$ (the $O(1/n)$ noise in sample correlations summed over $\sim M^2$ pairs). Collecting the three caps, $1/M_e := \text{Var}(A_{ij}) \approx 1/n + 1/M + \overline{\rho^2}$.

A.4 The Hadamard relation $\text{Var}(W_{ij}) = 2 \text{Var}(A_{ij})^2$

A_{ij} ($i \neq j$) is a normalized sum of M weakly dependent terms, so by a central limit theorem $A_{ij} \approx \mathcal{N}(0, 1/M_e)$. The AxA kernel is the centered Hadamard (elementwise) square of the additive GRM, $W_{ij} \propto A_{ij}^2 - \bar{A}^2$. For $X \sim \mathcal{N}(0, v)$, $\text{Var}(X^2) = 2v^2$, so $\text{Var}(W_{ij}) \propto \text{Var}(A_{ij}^2) = 2 \text{Var}(A_{ij})^2 = 2/M_e^2$; normalizing the kernel so the constant is unity gives $\text{Var}(W_{ij}) = 2/M_e^2$, hence $M_e = \sqrt{2/\text{Var}(W_{ij})}$.

A.5 Detection-regime N_{eff} and the standard error (11)

At small signal the off-diagonal estimator (10) is $\widehat{\sigma_{AA}^2} = (\sum_{i < j} W_{ij} y_i y_j) / (\sum_{i < j} W_{ij}^2)$. Under $y \sim \mathcal{N}(0, I)$ the cross-products $\{y_i y_j\}_{i < j}$ are uncorrelated with unit variance, so $\text{Var}(\text{numerator}) = \sum_{i < j} W_{ij}^2$ and

$$\text{Var}(\widehat{\sigma_{AA}^2}) = \frac{1}{\sum_{i < j} W_{ij}^2} \implies N_{\text{eff}} := \frac{2}{\text{Var}(\widehat{\sigma_{AA}^2})} = \sum_{i \neq j} W_{ij}^2 = (n^2 - n) \text{Var}(W_{ij}) \approx \frac{2n^2}{M_e^2}. \quad (17)$$

Thus $\text{SE}(h_{AA}^2) = \sqrt{2/N_{\text{eff}}} = M_e/n$, the off-diagonal estimator's $c = 1$. A full-information estimator (also using the diagonal y_i^2 , as REML does through its inverse-variance weighting) raises N_{eff} by a constant factor c^2 , giving $\text{SE}(h_{AA}^2) = M_e/(cN)$ as in (11); the value of c for the three-component model is read off the exact CRB in §6.